

Thommes et al. (1999, 2002) Ice Giant Formation Paradigm

Formation of Uranus and Neptune in the Jupiter-Saturn Region via Gravitational Scattering & Dynamical Friction

1. In Situ Accretion Crisis

- Standard growth time:
 $t_{\text{acc}} \propto \frac{a^{3.5}}{\Sigma_0}$
 - At $a = 19.2$ AU (Uranus): $t \sim 180$ Myr
 - At $a = 30.1$ AU (Neptune): $t \sim 850$ Myr
 - Nebula gas lifetime: $\tau_{\text{gas}} \approx 3 - 5$ Myr
- ⇒ Cannot capture observed
 $1 - 2 M_{\oplus}$ H/He envelope in situ!

2. Oligarchic Core Growth

- Cores grow between J & S:
 $a_{\text{init}} \in [5.5, 8.5]$ AU
- High solid surface density:
 $\Sigma_{\text{solid}} \approx 30 - 50 \text{ g/cm}^2$
- Fast accretion timescale:
 $t_{\text{acc}} \sim 1.5 - 3.5 \text{ Myr} < \tau_{\text{gas}}$
- Multiple $10 - 15 M_{\oplus}$ cores
readily form before gas dispersal.

3. Gas Giant Scattering

- Jupiter ($318 M_{\oplus}$) & Saturn ($95 M_{\oplus}$) undergo runaway gas accretion.
- Strong resonant perturbations destabilize proto-ice giants.
- Safronov number $\Theta_J \approx 10.39 \gg 1$
- Cores scattered onto eccentric outer orbits: $Q \sim 20 - 40$ AU, $e \sim 0.6$.

4. Dynamical Friction Damping & Planetary Decoupling in Outer Disk

- Massive Primordial Planetesimal Disk:
 $M_{\text{disk}} \approx 30 - 50 M_{\oplus}$, $r \in [10, 32]$ AU
- Chandrasekhar Dynamical Friction Drag:
 $\mathbf{F}_{\text{df}} = -\frac{4\pi G^2 M_{\text{core}}^2 \rho \ln \Lambda}{v^3} \mathbf{v}$
- Damping timescale: $\tau_{\text{damp}} \approx 1.0 - 3.0$ Myr
- Eccentricity Circularization: $\frac{de}{dt} = -\frac{e}{\tau_{\text{damp}}} \Rightarrow e \rightarrow 0.01 - 0.05$
- Perihelion Lifting Mechanism:
 $\frac{dq}{dt} \approx \frac{ae}{\tau_{\text{damp}}} [1 - 2e(1 - e)] > 0$
- Perihelia lift out of Jupiter/Saturn reach ($q > 15$ AU), freezing stable modern orbits at $a_U \approx 19.2$ AU, $a_N \approx 30.1$ AU.